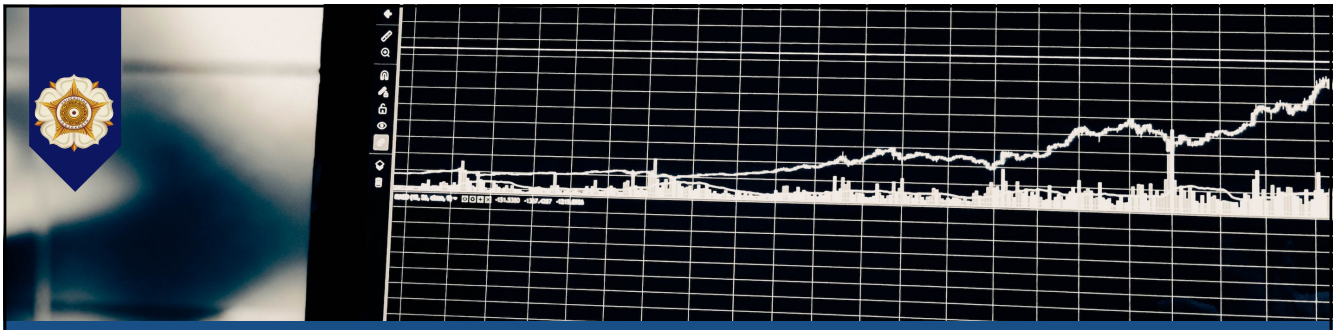


Introduction to Hypothesis Testing

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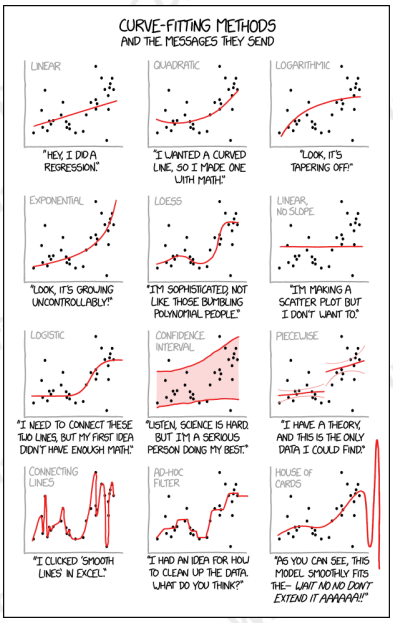
Version: March 2025

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1

Overview of today's lecture

- Experimental study and its variables
- Overview of hypothesis testing
- Hypothesis testing with large sample size
- Hypothesis testing with small sample size



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Part 1

EXPERIMENTAL STUDY AND ITS VARIABLES

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Types of quantitative research

Types of research	Focus	General claims	Typical methods
Descriptive	Describe a situation or a set of events	X is happening	Observations, field studies, focus groups, interviews
Relational	Identify relations between multiple variables	X is related to Y	Observations, field studies, focus groups, interviews
Experimental	Identify causes of a situation or a set of events	X is responsible for Y	**Controlled experiments**

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Experiment



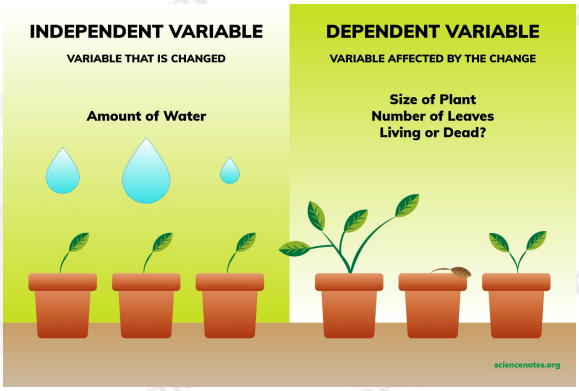
https://www.readandspell.com

- Researcher recruits 20 students, and assign them randomly into two groups
- 1st group: playing game in 3 months **(treatment group)**
  2nd group: not playing game **(control group)**
- Experiment: measure typing speed of those groups
- Conclusion : playing game improves the typing skills of the teenagers

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Independent and dependent variables

- Independent variables (IV) refer to the factors that the researchers are interested in studying or the possible "cause" of the change in the dependent variable.
  - IV is independent of a participant's behavior.
  - IV is usually the treatments or conditions that the researchers can control.
- Dependent variables (DV) refer to the outcome or effect that the researchers are interested in.
  - DV is dependent on a participant's behavior or the changes in the IVs
  - DV is usually the outcomes that the researchers need to measure.



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Features of true experiment

- Manipulates independent variables
- Hold all other variables constant (hence, controlled experiment)
- Measure any change in the dependent variables

Manipulation of independent variables → **Produces** → Change in dependent variables

IV: Game playing
Task: Ask participants to play / not to play game

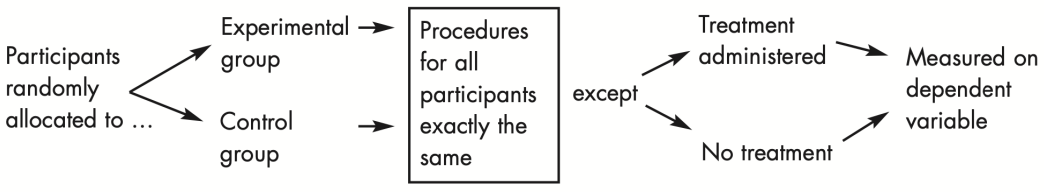
DV: Typing speed
Metrics: Words per minute (WPM)

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Features of true experiment

- Manipulates independent variables
- Hold all other variables constant (hence, controlled experiment)
- Measure any change in the dependent variables



Source: Coolican, Hugh. *Research methods and statistics in psychology*. Psychology press, 2017.

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Research hypothesis

- A scientific research (e.g. experiment) normally starts with a research hypothesis.
- A hypothesis: a precise idea that can be directly tested through an empirical investigation.
- A thesis: a tested hypothesis
- A theory: a statement that covers larger scope, which is developed from several empirical investigations



Hypothesis is an idea you can test
Buddy, Dinosaur Train

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Another scenario…

- You are proposing a new text entry technique for touchscreen smartphones based on artificial intelligence (gesture recognition and word prediction)
- You research goal is to achieve better entry performance than the conventional QWERTY soft keyboard (QSK)
- You decide to undertake an experimental research to evaluate your invention.

What are your hypothesis?



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From idea to hypothesis

The proposed user interface is better than the conventional user interface

The proposed finger gesture interface will be more likely yielding higher entry speed (in words per minute) on typing short message (less than 50 words) than the conventional QWERTY-based interface

Independent variable: smartphone input interfaces

Dependent variable: entry speed (words per minute)

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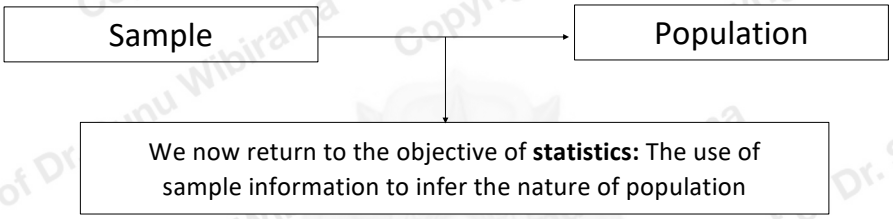
Part 2

OVERVIEW OF HYPOTHESIS TESTING

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Hypothesis testing



- Why do we need hypothesis testing?
  - When the population is large, we can only sample a sub-group of people from the entire population.
  - Hypothesis testing allows us to determine how confident we are that the results observed from the sampling population can be generalized to the entire population.
  - The use of each statistical testing highly depends on the design of experiment (e.g. repeated-measures / paired samples vs. between subject / independent samples) and normality of the data.

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Types of hypotheses in statistical point of view

- **Null Hypothesis** (H_0): typically states that there is no difference between experimental treatments.
- **Alternative Hypothesis** (H_1): a statement that is mutually exclusive with the null hypothesis.
- The goal of an experiment is to find statistical evidence **to refute or nullify the Null Hypothesis** in order to support the Alternative Hypothesis.

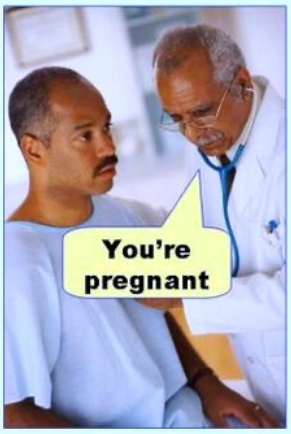
		What is true in the real world?	
		There is no effect (null = true)	There is an effect (null = false)
What conclusion is reached?	No effect	Correct conclusion (p = 1 - α) [True Negative]	Type II error (p = β) [False Negative]
	An effect	Type I error (p = α) [False Positive]	Correct conclusion (p = 1 - β) [True Positive]

Abbreviations: p: power; α: alpha or probability of a false positive result (i.e., Type I error); β: beta or probability of a false negative result (i.e., Type II error)

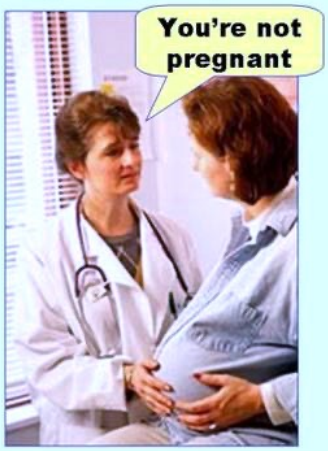
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Type I error
(false positive)



Type II error
(false negative)



Alpha = type 1 error
You believe that there is a "genuine"
effect of your interface on task completion time
but in reality, there is no effect

Beta = type 2 error
You believe that there is no "genuine"
effect of your interface on task completion time but in
reality, there is an effect.

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Important step of hypothesis testing

1. Define Null Hypothesis (H_0) and Alternative Hypothesis (H_a)
2. Define significant level (α)
3. Define statistical method / testing criteria
   (using Z-Distribution, t-Distribution, etc.)
4. Decision making: Accept or reject null hypothesis
5. Declare the final conclusion

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Step 1:

1. Define Null Hypothesis (H_0) and Alternative Hypothesis (H_a)
2. Define significant level (α)
3. Define statistical method / testing criteria (using Z-Distribution, t-Distribution, etc.)
4. Decision making: Accept or reject null hypothesis
5. Declare the final conclusion

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Null Hypothesis (H_0)

- H_0 states that:
  - Two or more variables are not related each other
  - No difference between two or more variable

Examples:

➢There is no relationship between rain fall and electricity connection
➢Monthly income is not related with preference of buying gadget
➢ There is no difference of GPA between activists and non activists

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Alternative hypothesis (H_a or H_1)

- H_a states that:
  - Two or more variables are closely related
  - there is difference between the 1st variable and the 2nd variable

Examples:

➢ Rain fall and electricity connection are closely related

➢Monthly income and preference of buying gadget are closely related

➢ There is significance difference of GPA between activists and non activists

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Step 2:

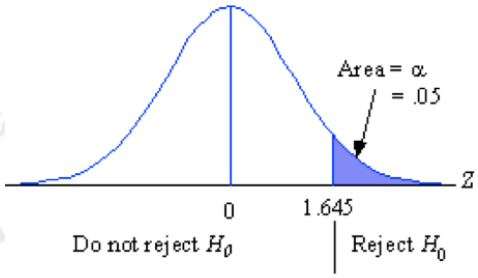
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Significant Level

- Define the limit of confidence level for your H_0
- Noted by *alpha* (α) symbol
- α = 1% , α = 5%, α = 10% , etc……



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Significant level

		What is true in the real world?	
		There is no effect (null = true)	There is an effect (null = false)
What conclusion is reached?	No effect	Correct conclusion (p = 1 - α) [True Negative]	Type II error (p = β) [False Negative]
	An effect	Type I error (p = α) [False Positive]	Correct conclusion (p = 1 - β) [True Positive]

Abbreviations: p: power; α: alpha or probability of a false positive result (i.e., Type I error); β: beta or probability of a false negative result (i.e., Type II error)

Probability of error type I occurred → significant level
Example: **α = 5%** → probability that we make error type I : 5%
(95% our decision tend to be correct)

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Step 3:

1. Define Null Hypothesis (H_0) and Alternative Hypothesis (H_a)
2. Define significant level (α)
3. Define statistical method / testing criteria (using Z-Distribution, t-Distribution, etc.)
4. Decision making: Accept or reject null hypothesis
5. Declare the final conclusion

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Statistical Testing

- Big Sample (n ≥ 30)

$$Z = \frac{Statistik_{sampling\ distribution} - Parameter_{population}}{\sigma_{sampling\ distribution}}$$

- Small Sample (n < 30)

$$t = \frac{Statistik_{sampling\ distribution} - Parameter_{population}}{\sigma_{sampling\ distribution}}$$

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Implementation

- Testing with only one mean:

$$Z_h = \frac{\bar{x} - \mu}{\sigma_{\bar{x}}} \qquad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

Standard deviation of sampling distribution / Standard Error

- Comparing means of two sampling distribution

$$Z_h = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

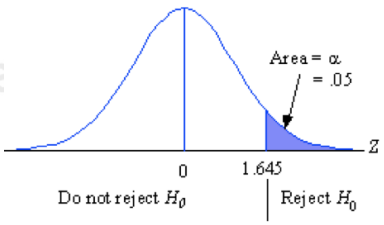
This is what normally goes in experimental study. Example: comparing average typing speed (in word per minute) between game player group and non game player group

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One-tail tests

- Right Tail



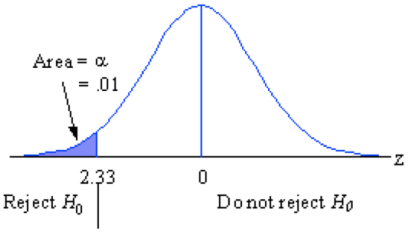
$H_0 : \mu = \mu_0$

$H_a : \mu > \mu_0$

Reject H_0 : $Z_h \geq Z_\alpha$; $t_h \geq t_{\alpha,n-1}$

Accept H_0 : $Z_h < Z_\alpha$; $t_h < t_{\alpha,n-1}$

- Left Tail



$H_0 : \mu = \mu_0$

$H_a : \mu < \mu_0$

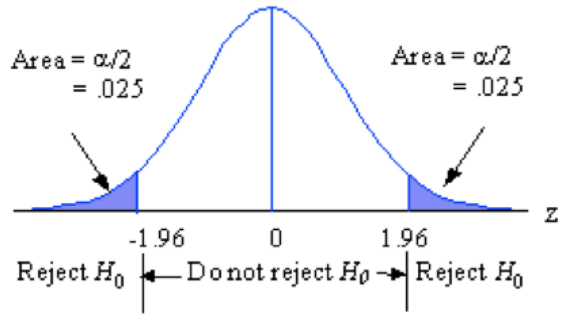
Reject H_0 : $Z_h \leq -Z_\alpha$; $t_h \leq -t_{\alpha,n-1}$

Accept H_0 : $Z_h > -Z_\alpha$; $t_h > -t_{\alpha,n-1}$

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Two Tail Tests



$H_0 : \mu = \mu_0$

$H_a : \mu \neq \mu_0$

Reject H_0 :

$Z_h \geq Z_{0.5\alpha}$

$Z_h < -Z_{0.5\alpha}$

$t_h \geq t_{0.5\alpha,n-1}$

$t_h < -t_{0.5\alpha,n-1}$

Accept H_0 :

$-Z_{0.5\alpha} < Z_h < Z_{0.5\alpha}$

$-t_{0.5\alpha} < Z_h < t_{0.5\alpha}$

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Step 4:

1. Define Null Hypothesis (H_0) and Alternative Hypothesis (H_a)
2. Define significant level (α)
3. Define statistical method / testing criteria (using Z-Distribution, t-Distribution, etc.)
4. Decision making: Accept or reject null hypothesis
5. Declare the final conclusion

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Step 5:

1. Define Null Hypothesis (H_0) and Alternative Hypothesis (H_a)
2. Define significant level (α)
3. Define statistical method / testing criteria (using Z-Distribution, t-Distribution, etc.)
4. Decision making: Accept or reject null hypothesis
5. Declare the final conclusion

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Final Decision

- Should be stated clearly
- Answering the problem

Examples:

➢It was true that no relationship between rain fall and electricity connection

➢Monthly income is proven not related with preference of buying gadget

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Part 3

HYPOTHESIS TESTING WITH LARGE SAMPLE SIZE

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Example 1:

- A survey result states that on average, DTETI UGM students spend Rp. 400,000 per month. A lecturer suspects that the average spending of all DTETI UGM students is not Rp. 400,000—it could be either lower or higher than that.
- To investigate, the lecturer takes a sample of 100 students and tests the hypothesis at a **5% significance level**. From the research conducted, it is found that the **average spending of the 100 students is Rp. 390,000 with a standard deviation of Rp. 25,000**.

  Determine whether the lecturer's suspicion is correct.

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Step 1:

1. Define Null Hypothesis (H_0) and Alternative Hypothesis (H_a)
2. Define significant level (α)
3. Define statistical method / testing criteria (using Z-Distribution, t-Distribution, etc.)
4. Decision making: Accept or reject null hypothesis
5. Declare the final conclusion

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Step 1: Define Null Hypothesis (H_0) and Alternative Hypothesis (H_a)

Mathematically, we can define the H_0 and H_a as follows:

- **H_0** : the suspicion is not true. **There is no difference** between average spending based on samples ($\bar{x}$) and average spending of population (μ). The population mean (μ) that will be calculated falls within the **confidence interval** (the acceptance region of H_0). Hence, it is true that μ = 400,000
- **H_a** : the suspicion is true. **There is difference** between average spending based on samples ($\bar{x}$) and average spending of population (μ). The average population spending (μ) estimated from samples spending ($\bar{x}$ = 390,000) falls outside the confidence interval. In other words, $\mu < 400{,}000$ or $\mu > 400{,}000$

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Step 2:

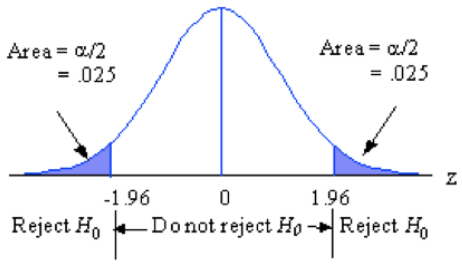
1. Define Null Hypothesis (H_0) and Alternative Hypothesis (H_a)
2. Define significant level (α)
3. Define statistical method / testing criteria (using Z-Distribution, t-Distribution, etc.)
4. Decision making: Accept or reject null hypothesis
5. Declare the final conclusion

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Step 2 : Define significant level (α)

- The data assumption: normal distribution.
- Check the Z distribution table. For a 5% significance level, alpha = 0.05. Therefore, the confidence coefficient is 95%.
- Since the hypothesis being tested is whether it is smaller OR larger, use a two-tail test. $z_{critical}$=-1.96 and 1.96



Confidence Coefficient		
$(1-\alpha)$	$\alpha/2$	$z_{\alpha/2}$
.90	.050	1.645
.95	.025	1.960
.98	.010	2.330
.99	.005	2.58

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Step 3:

1. Define Null Hypothesis (H_0) and Alternative Hypothesis (H_a)
2. Define significant level (α)
3. Define statistical method / testing criteria (using Z-Distribution, t-Distribution, etc.)
4. Decision making: Accept or reject null hypothesis
5. Declare the final conclusion

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Step 3: Define statistical method / testing criteria

- Since the sample size is large (n = 100), use the Z distribution (standardized normal distribution).

Theorem 8.2: **Central Limit Theorem:** If $\bar{X}$ is the mean of a random sample of size n taken from a population with mean μ and finite variance σ^2, then the limiting form of the distribution of

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}},$$

as $n \to \infty$, is the standard normal distribution $n(z; 0, 1)$.

$$z_{\text{calculated}} = \frac{\bar{x} - \mu}{\sigma_{\bar{x}}}$$

$$z_{\text{calculated}} = \frac{390{,}000 - 400{,}000}{25{,}000/\sqrt{100}} = \frac{-10{,}000}{2{,}500} = -4$$

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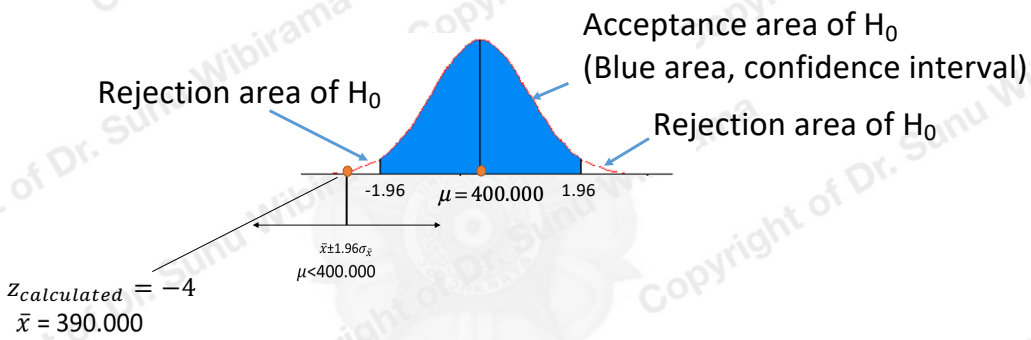
Step 4:

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2. Define significant level (α)
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4. Decision making: Accept or reject null hypothesis
5. Declare the final conclusion

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Step 4: Decision making: Accept or reject null hypothesis



Since $z_{calculated} < z_{critical} \rightarrow -4 < -1.96$,
in other words, the sample mean falls outside the confidence interval.
Therefore, the population mean is also less than 400,000.
Thus, H_0 is rejected.

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Step 5:

1. Define Null Hypothesis (H_0) and Alternative Hypothesis (H_a)
2. Define significant level (α)
3. Define statistical method / testing criteria (using Z-Distribution, t-Distribution, etc.)
4. Decision making: Accept or reject null hypothesis
5. Declare the final conclusion

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Step 5: Final conclusion

- **Tips:** Always go back to the original problem. Look at the question being asked.
- **The problem to be investigated:** define whether the average spending of all DTETI UGM students is Rp. 400,000.
- **CONCLUSION:** From the hypothesis testing results, it is concluded that H_0 is rejected, meaning the lecturer's assumption is correct — **the average spending of DTETI students is less than Rp. 400,000.**

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Part 4

HYPOTHESIS TESTING WITH SMALL SAMPLE SIZE

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Example 2:

- A survey result states that on average, DTETI UGM students spend Rp. 400,000 per month. A lecturer suspects that the average spending of all DTETI UGM students is not Rp. 400,000—it could be either lower or higher than that.
- To investigate, the lecturer takes **a sample of 25 students** and tests the hypothesis at a **5% significance level**. From the research conducted, it is found that the **average spending of the 25 students is Rp. 390,000 with a standard deviation of Rp. 25,000**.

  Determine whether the lecturer's suspicion is correct.

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Step 1: Define Null Hypothesis (H_0) and Alternative Hypothesis (H_a)

Mathematically, we can define the H_0 and H_a as follows:

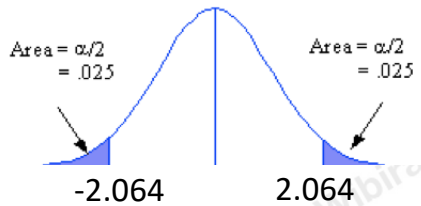
- **H_0** : the suspicion is not true. **There is no difference** between average spending based on samples ($\bar{x}$) and average spending of population (μ). The population mean (μ) that will be calculated falls within the **confidence interval** (the acceptance region of H_0). Hence, it is true that μ = 400,000
- **H_a** : the suspicion is true. **There is difference** between average spending based on samples ($\bar{x}$) and average spending of population (μ). The average population spending (μ) estimated from samples spending ($\bar{x}$ = 390,000) falls outside the confidence interval.
  In other words, μ < 400,000 or μ > 400,000

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Step 2 : Define significant level (α)

- The data assumption: small sample size.
- Check the student-t distribution table. For a 5% significance level, alpha (α) = 0.05. df=n-1, 25-1 = 24. α/2 = 0.025.
- Since the hypothesis being tested is whether it is smaller OR larger, use a two-tail test. $t_{critical}$=-2.064 and 2.064



df/p	0.40	0.25	0.10	0.05	0.025	0.01	0.005	0.0005
21	0.256580	0.686352	1.323188	1.720743	2.07961	2.51765	2.83136	3.8193
22	0256432	0.685805	1.321237	1.717144	2.07387	2.50832	2.81876	3.7921
23	0256297	0.685306	1.319460	1.713872	2.06866	2.49987	2.80734	3.7676
24	0.256173	0.684856	1.317836	1.710882	2.06390	2.49216	2.79694	3.7454
25	0.256060	0.684430	1.316345	1.708141	2.05954	2.48511	2.78744	3.7251
26	0.255955	0.684043	1.314972	1.705618	2.05553	2.47863	2.77871	3.7066
27	0.255858	0.683685	1.313703	1.703288	2.05183	2.47266	2.77068	3.6896
28	0.255768	0.683353	1.312527	1.701131	2.04841	2.46714	2.76326	3.6739
29	0.255684	0.683044	1.311434	1.699127	2.04523	2.46202	2.75639	3.6594
30	0.255605	0.682756	1.310415	1.697261	2.04227	2.45726	2.75000	3.6460
z	0.253347	0.674490	1.281552	1.644854	1.95996	2.32635	2.57583	3.2905
CI	———	———	80%	90%	95%	98%	99%	99.9%

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Step 3: Define statistical method / testing criteria

- Since the sample size is small (n = 25), use the t-distribution

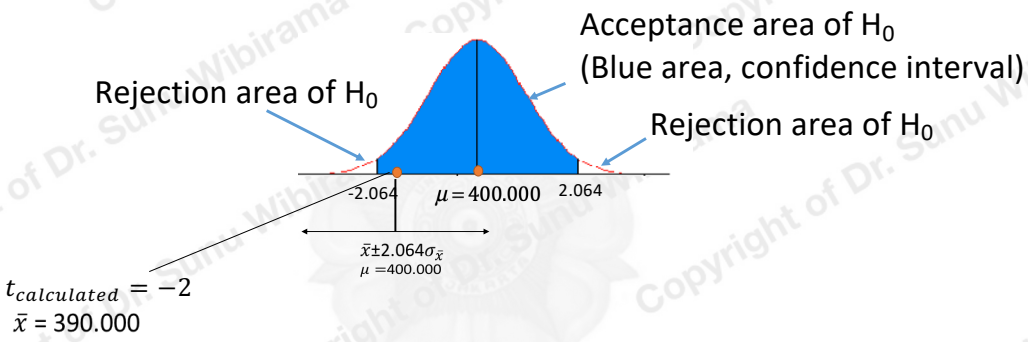
$$t_{\text{calculated}} = \frac{\bar{x} - \mu}{\sigma_{\bar{x}}}$$

$$t_{\text{calculated}} = \frac{390{,}000 - 400{,}000}{25{,}000/\sqrt{25}} = \frac{-10{,}000}{5{,}000} = -2$$

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Step 4: Decision making: Accept or reject null hypothesis



Since $t_{calculated} > t_{critical}$ → −2 > −2.064,
in other words, the sample mean falls inside the confidence interval.
Therefore, the population mean is 400,000. Thus, H_0 is accepted.

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Step 5: Final conclusion

- **Tips:** Always go back to the original problem. Look at the question being asked.
- **The problem to be investigated:** define whether the average spending of all DTETI UGM students is Rp. 400,000.
- **CONCLUSION:** From the hypothesis testing results, it is concluded that H_0 is accepted, meaning the lecturer's assumption is false — **the average spending of DTETI students is Rp. 400,000.**

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What about hypothesis testing with two means?

Introduction
I. Confidence Intervals and Estimates +
II. Hypothesis Testing for 1 Sample +
III. Hypothesis Testing for 2 Samples −
Hypothesis Testing for 2 Samples: Introduction
Hypothesis Testing: 2 Proportions
Hypothesis Testing: 2 Means (Independent Samples)
Hypothesis Testing: Matched Pairs
IV. Correlation and Regression +

INFERENCE FOR COMPARING 2 POPULATION MEANS (HT FOR 2 MEANS, INDEPENDENT SAMPLES)

More of the good stuff! We will need to know how to label the null and alternative hypothesis, calculate the test statistic, and then reach our conclusion using the critical value method or the p-value method.

The Test Statistic for a Test of 2 Means from Independent Samples:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

What the different symbols mean:

n_1 is the sample size for the first group

n_2 is the sample size for the second group

df, the degrees of freedom, is the smaller of $n_1 - 1$ and $n_2 - 1$

μ_1 is the **population mean** from the first group

https://pressbooks.pub/basicstatistics/chapter/hypothesis-testing-2-means/

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Hypothesis testing with two samples (Walpole's Ch. 10)

Two-Sample Pooled t-Test

For the two-sided hypothesis

$$H_0:\ \mu_1 = \mu_2,$$
$$H_1:\ \mu_1 \neq \mu_2,$$

we reject H_0 at significance level α when the computed t-statistic

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{s_p\sqrt{1/n_1 + 1/n_2}},$$

where

$$s_p^2 = \frac{s_1^2(n_1 - 1) + s_2^2(n_2 - 1)}{n_1 + n_2 - 2}$$

exceeds $t_{\alpha/2,n_1+n_2-2}$ or is less than $-t_{\alpha/2,n_1+n_2-2}$.

Recall from Chapter 9 that the degrees of freedom for the t-distribution are a result of pooling of information from the two samples to estimate σ^2. One-sided alternatives suggest one-sided critical regions, as one might expect. For example, for H_1: $\mu_1 - \mu_2 > d_0$, reject H_1: $\mu_1 - \mu_2 = d_0$ when $t > t_{\alpha,n_1+n_2-2}$.

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Hypothesis testing with two samples (Walpole's Ch. 10)

Example 10.6: An experiment was performed to compare the abrasive wear of two different laminated materials. Twelve pieces of material 1 were tested by exposing each piece to a machine measuring wear. Ten pieces of material 2 were similarly tested. In each case, the depth of wear was observed. The samples of material 1 gave an average (coded) wear of 85 units with a sample standard deviation of 4, while the samples of material 2 gave an average of 81 with a sample standard deviation of 5. Can we conclude at the 0.05 level of significance that the abrasive wear of material 1 exceeds that of material 2 by more than 2 units? Assume the populations to be approximately normal with equal variances.

***Solution*:** Let μ_1 and μ_2 represent the population means of the abrasive wear for material 1 and material 2, respectively.

1. H_0: $\mu_1 - \mu_2 = 2$.
2. H_1: $\mu_1 - \mu_2 > 2$.
3. $\alpha = 0.05$.
4. Critical region: $t > 1.725$, where $t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{s_p\sqrt{1/n_1 + 1/n_2}}$ with $v = 20$ degrees of freedom.
5. Computations:

$$\bar{x}_1 = 85, \quad s_1 = 4, \quad n_1 = 12,$$
$$\bar{x}_2 = 81, \quad s_2 = 5, \quad n_2 = 10.$$

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Hypothesis testing with two samples (Walpole's Ch. 10)

Hence

$$s_p = \sqrt{\frac{(11)(16) + (9)(25)}{12 + 10 - 2}} = 4.478,$$
$$t = \frac{(85 - 81) - 2}{4.478\sqrt{1/12 + 1/10}} = 1.04,$$
$$P = P(T > 1.04) \approx 0.16. \qquad \text{(See Table A.4.)}$$

6. Decision: Do not reject H_0. We are unable to conclude that the abrasive wear of material 1 exceeds that of material 2 by more than 2 units.

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Summary

- In this lecture, we learn about how to construct a hypothesis for an experimental study.
- There are five important steps of hypothesis testing. All steps should be conducted properly.
- To define rejection area, you should pay attention on the type of distribution that you use and the critical value.
- Your computed *z* value or *t* value will be compared with the critical value. Reject Null Hypothesis (H_0) if the computed value falls in the rejection area.
- Always state your final conclusion based on the initial problem that you want to solve.

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Midterm exam

- The duration of the examination is 120 minutes (2 hours).
- The use of books and online resources is strictly prohibited during the examination.
- Students is strongly suggested to bring a single A4-sized sheet of notes and a scientific calculator (mobile phones and other electronic devices are not permitted).
- Both the problem sheet and the answer sheet must be submitted upon completion of the exam.
- Students may answer the questions in any order (e.g., Question 1, 5, 4, 2, 3). However, each response must be written in full before proceeding to the next question, without interspersing answers to different questions. The clarity and organization of responses will be considered in the assessment.

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THANK YOU

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